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Energy Efficient Built Environment

A simulation-based evaluation of Energy demand and climate impact in two boxes: BBR vs Passive house standards

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A simulation-based evaluation of Energy demand and climate impact in two boxes:
BBR vs Passive house standards

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Abstract

In 2022, the construction and real estate sector in Sweden accounted for approximately 22.1 percent of the country´s total emission of greenhouse gases. To promote sustainability in the construction sector, various environmental requirements have been introduced, such as limit values on climate emissions, as well as encouraging innovations and new sustainable building methods, as well as promoting passive house standards.

The purpose of this study is to compare two different construction designs, BBR and passive boxes, and their impact on energy demand and climate changes. To investigate the purpose of this study, two boxes from ICE BOX Challenge were studied in two steps: IDA ICE was used to compare Energy demand for both boxes, and then the results from the energy simulations were used in One Click LCA in order to compare both boxes' climate impact.

The results of both IDA ICE and One click LCA showed that the passive box had lower values for Energy demand and climate impact, when compared to the BBR box.
The heating demand of the passive house was 108 kWh/m^2 while for BBR box 231 kWh/m^2. The total climate impact for passive box 14 240 kgCO_2e, while for BBR box 20 270 kgCO_2e. After comparing both boxes, the conclusion of this study shows that using the design and concept of passive house standards contribute to lower Energy demand and lower impact on the climate.

Keywords: Energy demand. climate impact. IDA ICE. One Click LCA. Sheep wool insulation. and ICE BOX Challenge.

Contents

Innehåll

1. Introduction 1
   1.1 Background 2
   1.2 Aims and Objectives 3
   1.3 Hypothesis 4
   1.4 Methodology 4
   1.5 Limitations 4

2 Literature Review and theory 5
   2.1 Passive house 5
   2.2 BBR 5
   2.3 Energy Performance 6
   2.3 Insulation types 7
   2.4 Heat Pump. heating demand and Peak heating load 7
   2.5 LCA 8
   2.6 IDA Indoor Climate and Energy 10

3 Materials and Methods 12
   3.1 Boxes Energy Simulation in IDA ICE 12
   3.2 Life Cycle Assessment (LCA) in One klick LCA 16
      3.2.1 Goal and Scope 17
      3.2.2 Life Cycle Inventory (LCI) 20
      3.2.3 Life Cycle Impact Assessment (LCIA) 21

4 Results 22
   4.1 IDA ICE- Simulation 22
      4.1.1 Energy Performance 22
      4.1.2 Thermal Comfort 23
   4.2 Climate Impact/ One klick LCA 23

5 Discussion 27
   5.1 Discussion of the results 27
      5.1.1 IDA ICE Simulation Results 27
      5.1.2 LCA Results 29
   5.2 Discussion of the method 30

6 Conclusions 32
   6.1Future Work 33

References 35

Appendices 39
   1. Drawing 39
      1.1 Passive Box Drawing 39
      1.2 BBR Box Drawing 40

Abbreviations

Abbreviation	Description
PH	Passive house box
IDA ICE	Implicit Differential Algebraic equations system solver - Indoor Climate and Energy
LCA	Life Cycle Assessment
LCC	Life Cycle Cost
BBR	The National Board of Housing. Building and Planning's building regulations
ASHRAE	American Society of Heating. Refrigeration and Air-Conditioning Engineers
ICT	Information and Communication Technology
EPD	Environmental Product Declaration
GWP	Global Warming Potential
LCIA	Life Cycle Impact Assessment
LCI	Life Cycle Inventory
A-drawing	Architectural drawing
K-drawing	Structural drawing
CAV	Constant Air Volume
COP	Coefficient of Performance
R-value	Thermal Resistance Value
DVUT	Dimensioning Winter Outdoor Temperature

Nomenclature

Symbol	Description	Unit
ρ	Density	kg/m^3
c	Specific heat capacity	J/(kg K)
λ	Thermal conductivity	W/(m K)
GWP	Global Warming Potential	$kgCO_2e/(m^2 yr)$
U-value	Thermal transmittance	$W/(m^2 K)$
DVUT	Dimensioning Winter Outdoor Temperature	°C
-	Climate impact of greenhouse gases	$kgCO_2e$
R-value	Thermal Resistance Value	m^2K/W

1. Introduction

The construction and real estate sector contributed in 2022 to the emissions of greenhouse gases with approximately 10.8 million tons of carbon dioxide equivalent, which is around 22.1 percent of Sweden´s total emissions of greenhouse gas [1].

The Ice Box Challenge is an international science experiment that aims to visually illustrate the difference between different buildings with various energy- efficient designs [2]. The experiment involves two small identical buildings: one constructed according to BBR requirements and the other constructed according to passive house requirements. The experiment is about measuring and comparing the ice remaining in both houses after being exposed to the same conditions outside for several weeks [3].

The Ice Box Challenge was first introduced in Sweden in 2025 in Borlänge. Students from Dalarna University, along with students from Luleå, Växjö, Östersund and Stockholm designed both houses for the experiment. During the experiment in Borlänge, students from Borlänge introduced a unique method of using locally sourced sheep wool as insulation material [4]. Sheep wool is made from a naturally produced fiber that requires less than 15% of the energy required for the production of glass fiber insulation [5].



Figure 1. Two buildings designed for Ice Box Challenge. the one according to BBR requirements and the one on the right according to passive house standards [2].

The results of the Ice Box Challenge are shown in the figure below, which shows that the block that was placed in the BBR box has melted significantly with visible water loss, while the other ice block appears to have retained more of its original volume and structure [3].



Figure 2 The results of the Ice Box Challenge in Borlänge 2025 [3].

1.1 Background

Passive houses have lower energy demand than conventional houses. The evidence presented by Jonsson and Brandt in the report *Passivehusen på* Oxtorget [6] suggests that passive houses differ in function and construction in comparison with conventional houses. The results of this report show that passive houses have an initial higher cost estimated at approximately 10-15% more than conventional houses. The design of passive houses can reduce Energy demand by 60% compared to conventional houses. The report shows that there are no disadvantages in passive houses from a performance perspective. The challenge identified in the study is the lack of awareness and experience within the construction sector in Sweden regarding buildings with passive house techniques [6].

Buildings that are not built according to passive house design account for a significant share of energy use, and the urgent need to reduce Energy demand in buildings is well recognized. Cederlund and Josefsson [7] claim in their report *Passiva flerbostadshus* that both the residential and service sectors account for approximately 40% of the total energy use in Sweden. The report discusses potential ways to save energy by approving improving the building´s energy efficiency. The design of passive houses minimizes heating demand and require less energy in comparison to conventional houses [7].

One of the advantages of passive houses is the long-term economic savings demonstrated by Ahmad and Vieglins in *Energi- och kostnadsjämförelse hos BBR och passiva flerbostadshus* [8]. The report investigates how energy performance of an existing apartment building in Stockholm, constructed in accordance with BBR, may have differed if the same apartment building was constructed in accordance with passive house standards outlined in FEBY12. A part of the study focuses on evaluating the life cycle costs (LCC) for the building with BBR and passive house criteria. The results of the report [8] show that the apartment building, constructed in accordance with passive house criteria, lowers Energy demand and offers higher thermal comfort indoors.

Beside implementing passive house standards, the choice of alternative insulation materials has additional environmental and functional benefits. The importance of choosing an

insulation material with lower climate impact is discussed in a report by Amer and Gultekin [9]. The goal of the study was to investigate the climate impact of passive houses, based on both Energy demand and required materials. The study compared a single passive house with three houses in three different Swedish cities, all using district heating systems. How much materials needed for the building envelope was also studied in the report, and the need for materials in the walls, floor, roof was investigated, and the Energy demand for each house in different climates, based on the passive house standard FEBY12. The results of that study indicated that the climate impact for the passive house was higher with 31.6% from materials than the other buildings. However, since the embodied emissions from materials remain constant over time, the passive house becomes more sustainable and energy efficient in the long terms. The Option of passive house becomes a better alternative after [9]:

- 12.5 years in Umeå
- 14.6 yeas in Borlänge
- 19.1 years in Malmö

The analysis of LCC showed that the cost for the existing house, designed according to BBR was 41.6 million SEK, while for passive house 40.5 million SEK over a period of 40 years. This means savings of 1.1 million SEK over a 40-year period. The conclusion of the study is that although a building designed according to the passive house standard requires higher initial material costs, the economic advantages are higher in the long term [9].

There are few studies which investigate the advantages and disadvantages of sheep wool as insulation in building, from the aspect of energy performance and climate impact. Kabun and Reinhold [10] report that the potential of using sheep-wool to enhance indoor air quality and reduce health risks in workplaces. Two cabins (2 m^3) were constructed, one insulated with sheep-wool based panels with renewable thermoplastic binder (PLA) and the other cabin insulated with mineral wool and polyurethane. The results of the study showed that sheep-wool panels offer better sound insulation performance than traditional materials (mineral wool). This study offers a different perspective by investigating new aspects, such as thermal comfort and climate impact, which results in a broader context [10].

To fill existing gaps, which were never investigated before about studying energy performance and climate impact for houses insulated with sheep wool, the earlier studies provide an important foundation by offering valuable information and methods.

1.2 Aims and Objectives

The purpose of the thesis is to study how the design of the two buildings used in the Ice Box Challenge in Borlänge, one following BBR standards and the other passive house standards, affect energy demand. thermal comfort and the climate.

Although the Ice Box Challenge has already offered knowledge of energy efficiency internationally, the goal of this study is to offer a detailed and comparative analysis of energy demand and climate changes as results of the building standards according to BBR and Passive house regulations.

The goal is to investigate both buildings' impact on the climate resulted according to the required energy for them.

The research questions used in this thesis are:

- How heating demand differs in both studied boxes?

- How does the climate impact differ in comparison between BBR box and passive box in the context of the Ice Box Challenge?

1.3 Hypothesis

In this study, two boxes with different construction standards are studied. One is built according to BBR-accommodations and the other built according to Passive house standards.

The concept of Passive house is to make a house more sustainable in long terms and encourage the use of recycled materials. Therefore, the results of this study are expected to show that the passive box has lower energy demand which also means lower climate impact. The passive box is also expected to be more well-insulated and therefore more sustainable due to the encouragement of natural materials.

1.4 Methodology

In order to investigate energy demand for each box, the program IDA ICE will be used for energy simulations. In IDA ICE, a weather file with historical values for the temperature is used. Furthermore, One Click LCA was used to investigate climate impact for each box, based on their energy demand and materials.

1.5 Limitations

Before conducting this study, there was an obstacle in the form of lack of theory as a ground for this study, when it comes to investigating energy demand and climate impact in houses that are insulated with sheep wool. Not having enough theory about Sheep wool results in uncertainty while discussing and comparing Sheep wool with another insulation material.

Another limitation of this study when it comes to the background is that a Life Cycle Cost (LCC) was not included and analyzed in this study. The focus was to compare energy consumption and climate impact between BBR box and Passive box.

Since this study is based on the ICE BOX Challenge, where internal loads were not taken into account, the cooling system in the ceiling was removed. Some simplification was choosing Energy power for a heat pump with a size of 1.5 kW, while IDA ICE was offering a heat pump with a size of 3 kW. The reason behind this was that a heat pump with a size of 3 kW was considered big for such a small area as the area for the boxes in this study. Other simplifications were made in order to conduct this study. For example, the assumption of no one would be using the box as a "house" and no furniture in the box which could have affected energy demand.

2 Literature Review and theory

2.1 Passive house

Passive house is a type of building standard that was developed in Germany in the 1990s. The concept of a passive house is to use methods which regulate required heating and cooling indoors, and optimize sunlight and shading. There are five main characteristics which need to be considered when designing a passive house [11].

- An effective insulation that minimizes heat loss and heat gain contributes to a high-performance envelope.
- High quality windows are essential for a minimized heat loss and high-performance type of windows.
- Thermal bridges contribute to heat loss due to materials with lower insulation property. Therefore, materials with low thermal bridges are prioritized.
- Uncontrolled air movements in and out of a building causes lower effectiveness of the insulation, therefore designing an air tight construction is essential.
- The use of heat recovery ventilation for supplying fresh air though mechanical ventilation is highlighted in passive houses.

The characteristics of passive houses contribute to lower energy demand, up to 90% less heating and cooling energy than conventional buildings [11] [12].

2.2 BBR

BBR has been commissioned by the government of Sweden to introduce limit values for the climate impact of buildings. The limit values cover construction stage A1-A5 and the purpose of these limit values is to improve knowledge and contribute to reduced climate impact of buildings. A BBR (Boverkets Byggregler) house is built according to BBR regulations and includes mandatory provisions which must be achieved in order to ensure safety energy efficiency and comfort [13].

The regulations include nine chapters [13].

- Introduction
- General rules for buildings
- Accessibility. dwelling design. room height and utility rooms
- Mechanical resistance and stability
- Safety in case of fire
- Hygiene. health and environment
- Protection against noise
- Safety in use
- Energy management

Starting from July 1, 2025, major changes to BBR regulations will be implemented, and affect large construction projects and smaller renovations [14].

The goal of these new changes is [15]:

- Higher requirements for the standard of insulation.
- Installation of energy efficient heating systems. such as solar panels
- More adapted use of electricity in order to reduce energy demand.
- Use of sustainable building materials.

Primary energy factor is used as a measurement of a building´s energy performance. The primary energy values can be calculated by multiplying the building´s energy demand by using weighting factors in order to calculate heating energy. Different energy sources have different weighting factors, for example, electricity has a factor of 1.8 and district heating has a factor of 0.7. When heating energy demand has been calculated, the value is then divided by a factor that is adjusted geographically. The final step is to divide the value by the building's area A_{temp} [16].

2.3 Energy Performance

Internal heat gains from occupants, lighting and ICT (Information and Communication Technology tools, such as computers) account for a major share of heat gain in office buildings. Today, the use of ICE has become more popular, even as devices become more efficient. Modern buildings are built well insulated, which causes the internal heat gain to cover a good amount of the building's heat needed during winter. Different studies investigating plug loads (such as computers) have a major impact on the building's sustainability and energy bills. However, ICT is slightly the official energy performance rating of a building. The impact of ICT is particularly relevant in very efficient buildings with A+ or A++ labels [17].

Energy demand in modern, highly insulated houses, such as passive houses, is influenced by internal heat gains from occupants, lighting and equipment, as well as the house´s location, orientation, and facade design. Internal heat gains have a major influence on overall energy use [18] [19]. In passive houses, extensive glazing can create challenges related to solar heat gain. Passive houses are designed to have minimized heating demand through superior insulation, airtight construction and efficient heat recover ventilation. Passive houses can still be vulnerable to overheating during warmer climates due to intense solar radiation. Overheating increases the demand for mechanical cooling, which conflicts with the goal of the passive house idea of reducing operational energy demand. In order to reduce radiant heat exposure, shading strategies become essential in passive houses with large glazed surfaces [18][19].

A building´s energy demand is influenced by various factors, such as its architectural layout, building components, and material choices. Certain design decisions influence heating demand during winter [20].

Another important factor for heat demand is the shape factor defined as the ratio of a building's envelope area to its volume. Buildings with higher shape factors contribute to more heat loss and therefore require more energy. The shape factor can account for 10-20% of the final energy demand, but the influence decreases during warmer climates when the outdoor temperature is around 11-14 °C [21].

There has been a noticeable decline in energy use required for hot water and heating in Swedish small houses. In 2024, the average use of energy for heating and hot water (excluding electricity) for small houses was 90.5 kWh/m²[22]. Houses built in 2021 or later have energy demand of around 39 kWh/m² for heating and hot water, while houses built 1940 or earlier averaged about 110 kWh/m². The difference in energy use is due to modern construction practices, such as better insulation, improved building envelopes, updated heating systems and efficient windows [22].

2.3 Insulation types

There are different factors which can contribute to energy efficiency and sustainability, such as the choice of insulation.

Sheep wool is a natural product that can be used to insulate buildings. Sheep wool insulation is made from the wool of sheep. This natural resource comes from the annual shearing of sheep, where the wool is then carefully cleaned, processed and made into insulation mats, rolls or other products for insulation. Sheep wool is a particularly sustainable raw material, as it is fully biodegradable and can be produced in a continuous cycle without causing any damage [23]. There are several advantages to using sheep wool. It provides good thermal insulation, with a thermal conductivity of 0.035–0.04 W/(m K) [23] for different building elements, and good sound insulation, as well as moisture control, thanks to its natural ability to absorb and release it without losing its insulating effect. In addition, sheep wool is fire resistant and provides excellent protection against both heat and cold, making it a practical and safe choice for insulation [23].

Sheep wool is not always available in sufficient quantities or in different forms. This can result in longer waiting times for deliveries or a smaller variety of products. In general, sheep wool costs more than traditional insulation materials such as mineral wool (glass wool. rock wool) or cellulose. The cost reflects the greater work required for production and processing, as well as the limited availability [23].

Mineral wool is an example of an insulation that is common in buildings and differs from sheep wool. Mineral wool is widely used due to its high R-value, and good acoustic blocking and superior fire- resistance. The R-value refers to thermal resistance, that is, the ability of a material to resist heat transfer. A higher R-value means that the material provides better insulation. Mineral wool refers to two different types of insulation materials [24]:

Glass wool is made by melting glass, often recycled, and spinning it into thin threads that are then pressed into insulation boards or mats. This type is light and easy to install, and is usually the most cost-effective choice. For this reason, glass wool is widely used for insulating walls, ceilings and floors in ordinary homes. It has good properties for insulating against both heat and sound but is more sensitive to moisture. If the material gets wet, its capacity to insulate decreases [24].

Stone wool, on the other hand, is made by melting volcanic rock, such as basalt, which results in a heavier and denser material. This provides increased strength and makes it significantly better at soundproofing and fire safety, Stone wool is non-combustible and can withstand temperatures of up to about 1000 degrees, which makes it extra valuable in buildings where fire safety is of great importance. Unlike glass wool, it is not affected by moisture in the same way, which makes it more stable in damp areas such as basements. On the other hand, stone wool is heavier and requires more work, and it is also more expensive than glass wool [24].

2.4 Heat Pump. heating demand and Peak heating load

The size of an air-to-air heat pump can be determined by the floor area and the unit´s heat capacity. A normal house requires 40-50 W of heating power per square meter and around 70 W of cooling per square meter, assuming that the house has standard ceiling height, good insulation, and modern windows [25].

The size of a heat pump has an effect on energy efficiency. An undersized pump has to run at high load for long periods, which results in increased energy use. An oversized

pump is assumed to be more advantageous because it can operate at lower power with less effort [26].

There are different factors that need to be taken into account in order to determine the required heating capacity of an air-to-air heat pump:

The size of heated floor area:

- 80-120 m^2 requires around 5-6 kW
- 120-160 m^2 requires around 6-7 kW
- 160-200 m^2 requires around 7-8.5 kW [27].

The level of insulation and building age:

- Houses built after 2010 have very good insulation and generally lower heating demand.
- Houses built between 1980 and 2000 have normal insulation.
- Houses built before 1980 have high heating demand and therefore require an air-to- water pump or another heat pump with high capacity [27].

Climate zone:

- Southern Sweden has a milder winter climate and therefore a heat pump with lower capacity.
- Central Sweden has moderate winters climate and therefore requires a heat pump with standard capacity.
- Northern Sweden has a cold winters climate and therefore requires a heat pump with higher capacity [27].

Heat losses of houses can be estimated by construction period/insulation standard:

- Older houses, built before 1980 without added insulation have a typical heat loss between 120-180 W/m^2.
- Renovated houses, between 1980-2000. have a typical heat loss between 70-100 W/m^2.
- Houses built to 2000-2010 standards have a typical heat loss between 45-65 W/ m^2
- Modern houses, built after 2016. have a typical heat loss between 25-40 W/ m^2. [28].

2.5 LCA

Life Cycle Assessment (LCA) is a structured method used to evaluate the environmental impacts associated with a product or a building throughout its entire life cycle from raw material extraction and manufacturing to transport, construction use, maintenance, and end-of-life processes such as recycling or waste management. Conducting an LCA provides a holistic understanding of environmental performance and supports informed decision-making regarding material choices and construction strategies [29][30].

The climate impact of construction materials and building systems can be assessed at an early design stage using tools such as One Click LCA. By performing a life cycle assessment, the tool calculates the Global Warming Potential (GWP) and uses verified Environmental Product Declarations (EPD) to give a more accurate representation of the building's total climate impact [29][30].

Global Warming Potential (GWP) is a common metric used to express the climate impact of greenhouse gases relative to carbon dioxide. It quantifies how much a gas contributes to global warming over a specific period. The most widely used timeframe is GWP100, which

reflects impacts over 100 years, although shorter alternatives exist such as GWP20, which considers a 20-year perspective [29][30].

A life Cycle Assessment is about different periods of a building´s lifetime. To study every phase of a building´s lifetime, there are different stages in LCA:

- A1: Raw material extraction and processing phase
- A2: Transport to the manufacturer
- A3: Manufacturing
- A4: Transport to the building site
- A5: Installation into the building

LCA consists of more stages to study a building´s impact:

- B1-B9: Use stage plus operational energy and water use
- C1-C4: End of life stage
- D: Reuse, recovery and/or recycling potentials.

LCA provides a fundamental foundation to compare the environmental impact of the studied boxes with different construction designs and materials. LCA is based on international standards to outline its process, such as ISO 14040 [29][30].

ISO 14040 is international standards which outline the process for conducting LCA, and consists of four phases [29][30]:

Goal and scope phase: This phase involves defining and the scope of study, including the clarification of the study's purpose. The system boundaries are to be clearly defined in order to decide which stages of the life cycle to study.

Life Cycle Inventory (LCI): This phase focuses on data that is relevant for the LCA process, materials used, and energy performance, and emissions of the investigated building are of importance for this phase. This type of data can be collected from design documents, such as drawings, and database and literature reviews which are done earlier in a study.

Life Cycle Impact Assessment (LCIA): This phase focuses on using the data from LCI to calculate environmental impact of specific materials and the whole building in general. This provides a foundation for understanding how a building affects the environment throughout its whole life cycle, and when comparing different buildings and their climate impact.

Interpretation: This phase focuses on analyzing the results of LCIA to understand which factors contribute to the environmental impact. Then, a sensitivity analysis can be performed to investigate how changes in LCI or making assumptions that can affect the results.

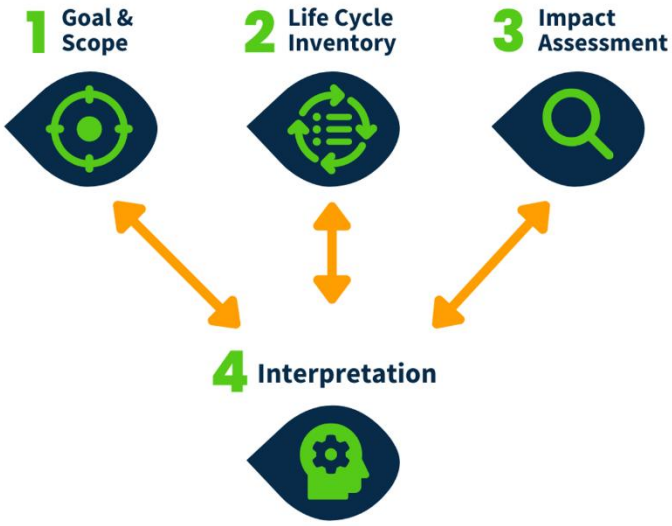


Figure 3. Phases of a Life Cycle Assessment (LCA) [30].

2.6 IDA Indoor Climate and Energy

IDA ICE was first introduced in 1995 in Sweden by EQUA Simulation AB, with financial support from the state and companies in the construction and real estate sector [31].

IDA ICE is a program for dynamic energy simulation and indoor climate calculation that can simulate buildings at the room level, often called zones. The program analyzes heat and mass transport between these zones and to the surrounding weather conditions. IDA ICE contains models for convection, conduction, radiation, air flows, etc. In addition to the annual energy requirement, power requirement, solar heat loads, sunlight. CO_2 concentration and thermal comfort can also be calculated. When a model is created, the program compiles a comprehensive system of equations consisting of nonlinear ordinary differential equations for all processes that occur in the building. The calculation time increases largely linearly with the size of the problem, which makes it recommended in the manual to combine several identical objects, such as windows, into one unit. For each zone that is drawn. about 2000 variables are added to the equation system [31].

IDA ICE is offered in three levels [32]:

- Wizard

The simplest level where users can create a zone and run a simulation or transfer the information to the standard level. The first version is called IDA Room wizard and the latest is called IDA ESBO. Both versions are part of the software and IDA Room wizard is also available for free on the internet.

- Standard

At this level, the user can enter geometry, materials, loads, schedules and more annual energy needs. power needs and thermal climate can be simulated here. Although a lot can be done at this level, it is user-friendly and there is good help available, such as step-by-step instructions.

- Advanced

In the advanced level, the user has the option to build the model using a block diagram instead of dialog boxes. Here the calculation data code can be reviewed and in the Expert version there is the option to edit and write your own code. All variables in the program can be logged for deeper studies [33].

IDA ICE is validated and certified according to ASHRAE 140. CEN Standard EN 15255 and 15265. CEN Standard EN 1379, where simulations of apartments with glazed balconies have been carried out, analyses of heat pumps, calibration of models based on measurement data and studies of CO_2 concentrations, among others [33]

3 Materials and Methods

To carry out the work in this study both IDA ICE and OneClickLCA, the implementation phase has been divided into two sections. The calculations of energy demand for each of the boxes have been carried out using the IDA ICE Standard version 5-1 program and global warming have been analyzed with One click LCA. Below is how the study has been carried out in its entirety and which calculations, comparisons and measurement results have been applied.

3.1 Boxes Energy Simulation in IDA ICE

In IDA ICE, a 3D representation of the boxes has been created to mimic the real construction as closely as possible, which includes aspects such as building technology, energy systems and heating solutions. The program uses climate information from among others, the database of the American Society of Heating, Refrigeration and Air-Conditioning Engineers (ASHRAE) to model the building's energy demand over a whole year. For this purpose, climate data for Borlänge was downloaded from that source and used in General for both boxes.

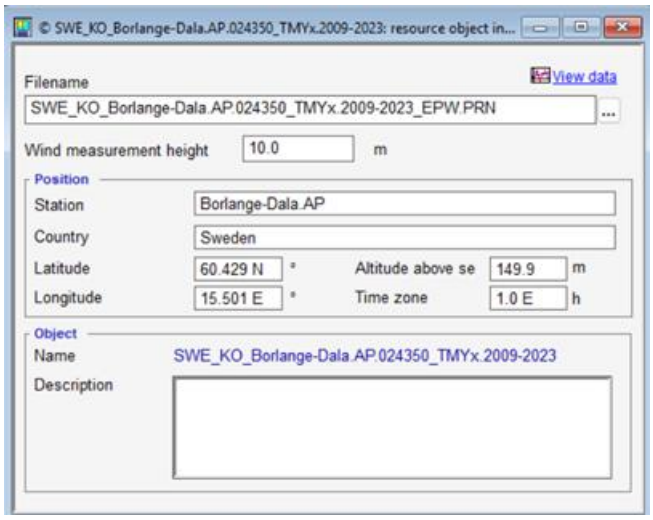


Figure 4. climate data for Borlänge.

Construction drawings have been obtained from Dalarna University. The A-drawing indicates dimensions while the K-drawing describes the construction and materials used for the roof, walls, and floors in all extensions (see Appendix A).

The next step was to specify dimensions in the floorplan button for both boxes, and the table 1 below provides more information about this.

Table 1. Describe the outer dimensions for both boxes.

Size	**Passivehouse box**	**BBR Box**
Width (mm)	3 244	2 304
Length (mm)	3 619	3 554
Height to ceiling (mm)	2 795	2 883
Floor area (m^2)	11.74	8.19

In passive houses, sheep wool was used as insulation material, while BBR houses used rock wool instead. In IDA ICE, there was no insulation for sheep wool which means that it must be defined as a new layer in in IDA ICE and therefore can be calculated according to the information presented below in table 2 and equations 1-5.

Wood in beams 45 mm wide with insulation placed in between with CC600 [34]

$$Woodproportion = \frac{d_{wood}}{cc} = \frac{45}{600} = 0.075 = 7.5\% \quad \text{Equation 1}$$

$$Sheep\ Wool\ proportion = \frac{d_{insulation}}{cc} = 1 - \frac{d_{wood}}{cc} = 1 - 0.075 = 0.925\% \quad \text{Equation 2}$$

Where:
d=Thickness of material [m]
cc=centre-to-centre studs

Table 2. Input data as described in Sheep´s Wool Insulation (Isolena) [35].

Parameter	Timber	Insulation	unit
Content	0.075	0.925	%
Density	450	20	kg/ m^3
Specific heat capacity	1600	1760	J/(kg K)
Thermal conductivity	0.12	0.034	W/(m K)

For both the Wood +insulation used the below equations [34]
Thermal conductivity (λ)

$$\lambda = \frac{d_{wood}}{cc} \times \lambda_{wood} + \frac{d_{insulation}}{cc} \times \lambda_{insulation} = (0.075 \times 0.12) + (0.925 \times 0.034) = 0.04045\ W/(m\ K) \quad \text{Equation 3}$$

Where:
λ =Thermal conductivity [W/m.K]
d =Thickness of material [m]
cc =centre-to-centre studs

For density used the below equations [36]

$$\rho = \frac{d_{wood}}{cc} \times \rho_{wood} + \frac{d_{insulation}}{cc} \times \rho_{insulation} = (0.075 \times 450) + (0.925 \times 20) = 52.25\ kg/m^3 \quad \text{Equation 4}$$

Where:
ρ = Density of the whole layer (kg/m3).
ρ_{wood}=Density of the timber stud (kg/m3)
$\rho_{insulation}$=Density of the insulation material (kg/m3)
d =Thickness of material [m]
cc= centre-to-centre studs

Specific heat capacity used the below equations [37]

$$C = \frac{\frac{d_{wood}}{cc} \times \rho_{wood} \times C_{wood} + \frac{d_{insulation}}{cc} \times \rho_{insulation} \times C_{insulation}}{\frac{d_{wood}}{cc} \times \rho_{wood} + \frac{d_{insulation}}{cc} \times \rho_{insulation}}$$ Equation 5

$$C = \frac{(0.075 \times 1600 \times 450) + (0.925 \times 1760 \times 20)}{(0.075 \times 1600) + (0.925 \times 1760)} = 49.7\ J/\ (kg\ K)$$

Where:
C = Specific heat capacity
ϱ_{wood}=Density of the timber stud (kg/m3)
$\varrho_{insulation}$=Density of the insulation material (kg/m3)
d =Thickness of material (m)
cc= centre-to-centre studs

Then specified walls, windows, floors and ceilings by selecting the surface in the floorplan and then going to advanced to fill in materials for the construction model. All material as used in the both boxes and thickness presented below in table 3.

After defining the building constructions, Ideal Heater and Ideal Cooler were assigned to the room units in IDA ICE to calculate the thermal energy demand of both boxes. The heating setpoint temperature was set to 21 °C and the cooling setpoint temperature was set to 26 °C. Since ideal systems were used, the heat source and cooling source were not modelled as real technical systems such as a heat pump or district heating, but as ideal loads. Therefore, the reported results represent the building's heating and cooling demand, not the electricity for heat pump operation. Then in the simulation button run heating design and cooling design.

Peak heating demand was determined in IDA ICE using the heating design function. The calculation was based on DVUT, which is the dimensioning winter outdoor temperature used to estimate the maximum heating demand of the boxes under cold outdoor conditions.

When it got the peak heat rate required for both boxes, it removed Ideal Heating and Ideal Cooler and added only the Water radiator in the room units since the energy requirement for cooling is non-existent. The value of heat rate and dimensions for water radiators are shown in table 4.

Table 3. Construction as used in the boxes and their thickness and area.

Building component	**Passivhaus box**	Thickness	Area	**BBR BOX**	Thickness	Area
WALL	-12.5 mm OSB -45 mm insulation(wollsheep) + 45x45 mm studs cc600. -0.2 mm vapor barrier -R400s masonite beams + 380 mm insulation -0.2 mm airtight membrane -28 mm airgap +28x70 mm studs cc600 -44 mm standing facade material (lockpanel)	510 mm	8.5 m^2	-10 mm OSB -0.2 mm vapor barrier -45x195 mm c24 joint+195 mm insulation -0.2 mm Wind protection -28 mm air gap+28x70 mm studs cc600 -44mm cover panel	277 mm	10.7 m^2
Window (facing North)	3 pan glazing		0.38 m^2	3 pan glazing		0.38 m^2
FLOOR	-0.2 mm vapor barrier -12.5 OSB -H450s masonite beams + 405 mm insulation. -12.5 OSB	430 mm	11.74 m^2	0.2 mm vapor barrier 10 mm OSB 45x220 mm C24 studs +220 mm insulation. 10 mm OSB	240 mm	8.19 m^2
ROOF	-12.5 OSB -0.2 mm vapor barrier -H450s + 405 mm insulation -Airtight membrane	418 mm	11.74 m^2	-20 mm Wood -28x70 mm air gap -45x200 C24 Studs + 200 mm insulation 0.2 mm vapor barrier 10 mm OSB	258 mm	8.19 m^2

Table 4. Describes the heat rate and dimensions for the used radiators for both boxes.

Parameter	Passivehouse Box	BBR Box
Dimension (m^2)	0.48	0.48
Heat rate (W)	210	370

After this, the energy system was determined by clicking on the General tab, Plant. It set up an Ambient air to water heat pump to cover the heating needs of the buildings. The heat pump was configured with a nominal heat output of 1.5 kW and a coefficient of performance (COP) of 3.0 under standard conditions and the same source set for both boxes [38].

The ventilation system was modelled as a central air handling unit using a Constant Air Volume (CAV) system. The supply and return airflows were set to 0.35 $l/(s \cdot m^2)$ for each zone, in accordance with BBR requirements for residential ventilation [39]. Infiltration was considered through the zone leakage area in IDA ICE. The model included a leak area of $6.01*10^{-4}$ m^2, while no additional constant in/exfiltration flow was added.

After defining the energy system, ventilation system and water radiator, the energy simulation and overheating analysis were activated in IDA ICE. The overheating analysis was used to evaluate the summer peak indoor temperature and thermal comfort.

After the IDA ICE simulation, the annual electricity demand for operating the heat pump was exported and used as input data in One Click LCA. The operational energy use was assessed over a reference study period of 30 years. The heating source considered in the LCA was an electricity-driven heat pump. Therefore, the climate impact was calculated based on the electricity demand required to operate the heat pump for both boxes.

3.2 Life Cycle Assessment (LCA) in One klick LCA

The life cycle assessment (LCA) has been carried out according to the principles of ISO 14040. The work includes the four main steps of an LCA: (1) Goal and Scope, (2) Life Cycle Inventory, (3) Life Cycle Impact Assessment and (4) Interpretation of Results.

The first component specifies the purpose and scope. This includes clarifying the goal of the specific life cycle assessment, the functional unit. system boundaries, as well as assumptions and possible limitations. The second component conducts a life cycle inventory. The third component deals with the assessment of life cycle impacts, where information from the inventory analysis is converted into a value for potential environmental impact. The fourth and final component interprets the collected results.

The OneClick LCA software was used to carry out this life cycle assessment. The analysis followed the standard EN 15804+A2 [40]. The life cycle inventory was produced by modeling the building in question in IDA ICE. from where all measurements were taken. These were entered manually into OneClick LCA, which was then able to perform the life cycle impact assessment. The building's Energy demand was calculated by IDA ICE. The first life cycle assessment for each unit was carried out using an attribution method, which means that the total environmental impact was assessed based on the materials and energy use of the building under the current conditions.

3.2.1 Goal and Scope

The objective of this study was to analyze the climate impact (Global Warming Potential. GWP) arising from two different boxes (BBR box and Passive house box) that participated in the Ice Box Challenge, which illustrate different methods of construction and insulation. The study aims to determine which of the two alternatives causes a lower climate impact during both the manufacturing and construction processes and to raise awareness of how the choice of materials affects the environmental impact of buildings.

Functional Unit

The functional unit was defined as 1 m² of the building envelope surface in order to enable a fair comparison between the two boxes, since their overall dimensions and total areas are different. By using 1 m², the environmental impact and thermal performance can be compared on an equivalent basis independent of the total size of each box. All simulations and calculations performed are based on a predefined period of 30 years.

System boundaries and method selection

The analysis includes all life cycle modules- in accordance with the EN 15804 +A2 standard, which means that everything from raw material extraction, transport and manufacturing to construction is included.

Materials that constitute less than 1% of the total weight, or that are considered to have a very low environmental impact, have been excluded according to the cut-off principle in ISO 14044.

Geographical and temporal delimitation

The geographical scope covers Borlänge, where the Ice Box Challenge took place, and that the results are based on Swedish conditions regarding transport and production. The time frame is 2025, which corresponds to the current state of technology and the available EPD data at the time of the study.

Data quality

The quality of the input data was assessed based on the material and energy data used in the study. To improve transparency, Table 5 shows the quantities of the main materials included in the life cycle assessment for both boxes. The material quantities were taken from the building model (Appendix 1) and entered manually into One Click LCA, where the environmental impacts were then calculated.

Table 5 Material quantities for wall included in the life cycle assessment for the two boxes.

Material	**Type**	**Quantity BBR**	**Quantity Passive House**	**Unit**	**Supplier (Sweden)**
Planed timber (spruce/pine). 481 kg/m³	Timber frame / studs	6.16	29.91	m³	Beijer
SBS bitumen vapor barrier membrane (YEP 3500)	Vapor barrier	16.4	37.17	m²	Nordic Waterproofing AB
Electric horizontal radiant panel. 1000W	Electric radiator	1	1	unit	Purmo
Softwood plywood. 480 kg/m³	Plywood sheet	0.32	0.175	m³	Beijer
Sheep wool insulation	Insulation	-	58.55	m²	Thermafleece
Stone wool insulation	Insulation	25.44	-	m²	Beijer
Charred wood cladding. 15 mm	External cladding	31.6	37.17	m²	Moelven
Wooden frame window. triple glazed. side-hung. 38.2 kg/m2	Window	0.38	0.38	m²	Elitfönster

Table 5 presents the material quantities included in the life cycle assessment for the wall assemblies of the two boxes. The corresponding material quantities for the floor and roof constructions are presented in tables 6 and 7.

Table 6 Materials and resources used. and their quantities and units in the floor construction

Material	**Type**	**Quantity (BBR)**	**Quantity (Passive House)**	**Unit**	**Supplier (Sweden)**
SBS bitumen vapor barrier membrane	Membrane	8.2	11.74	m^2	Nordic Waterproofing AB
Stone wool (mineral wool) insulation	Insulation	8.2	–	m^2	Beijer
Sheep wool insulation	Insulation	–	4.4	m^3	Thermafleece
Wooden I-beam H300 with OSB core	Structural beam	–	0.41	m	Masonite Beams AB
Wooden beam 47×47 mm (300 mm)	Structural beam	1.8	–	m^3	Masonite Beams AB
Plywood 460 kg/m^3	Sheet material	0.028	0.237	m^3	Beijer
Plywood (generic. 620 kg/m^3)	Sheet material	0.028	0.175	m^3	Beijer
OSB / Cross-laminated sheet in I-beam	Structural sheet	–	Included in H-beam	–	Masonite Beams AB

Table 7 Materials and resources used. and their quantities and units in the roof

Material	Type	Quantity (BBR)	Quantity (Passive House)	Unit	Supplier (Sweden)
Wooden H-type I-beam with OSB core (H300)	Structural beam	–	0.38	m	Masonite Beams AB
Stone wool (mineral wool) insulation	Insulation	8.2	–	m^2	Beijer
Sheep wool insulation	Insulation	–	11.74	m^2	Thermafleece
Plywood. generic 620 kg/m^3	Sheet material	0.082	0.0189	m^3	Beijer
Charred wood cladding. 15 mm	Cladding	8.2	11.74	m^2	Moelven
Bitumen roof waterproofing layer (5.3 kg/m^2)	Waterproofing	8.2	11.74	m^2	Trebolit / Mataki
Wooden beam I-shaped 47×47 mm. 300 mm	Structural beam	1.64	–	m^3	Masonite Beams AB

3.2.2 Life Cycle Inventory (LCI)

The inventory phase included the collection and quantification of all relevant input and output flows. In this study, the assessment was based on the constructions used in the two boxes, the quantities of the included materials, and the annual electricity demand for heat pump operation obtained from the IDA ICE simulations. Table 3 presents the constructions used in the boxes and their dimensions. Tables 5, 6 and 7 present the material quantities included in the life cycle assessment for the two boxes. These tables show the input data used in both IDA ICE and One Click LCA.

The second step was to enter the building materials presented in Table 3 for both boxes, including walls, ceilings, floors, frames and windows. Materials from the same suppliers were used where possible to ensure consistency in the comparison. Tables 5–7 present the materials and resources used in the wall, roof and floor constructions for the BBR box and the Passive house box. Material specifications, densities and quantities were based on Swedish suppliers and databases to ensure that the assessment reflects materials commonly available on the Swedish market.

After entering the material data, the annual electricity demand for heat pump operation was added to One Click LCA. The values used were 430 kWh for the Passive house box and 635 kWh for the BBR box, obtained from IDA ICE. In One Click LCA, the electricity dataset "Electricity, consumption mix w/o renewables, Sweden" was selected as the energy source for both boxes. The operational energy use was assessed over a reference study period of 30 years.

When all input data had been entered, the calculation was performed in One Click LCA. The software calculated the climate impact of the materials and operational energy use. The results were then used to analyses and compare the global warming potential of the two boxes.

The life cycle stages included in this study were A1–A5, B4–B6 and C2–C4, following the structure used in One Click LCA. The product stage A1–A3 includes raw material extraction, transport to the manufacturer and manufacturing of the building products. Stage A4 includes transport of materials to the construction site, while A5 includes construction and installation processes.

In the use stage, B4 includes replacement of building components during the reference study period, B5 includes refurbishment, and B6 includes operational energy use. In this study, B6 was based on the annual electricity demand for heat pump operation obtained from IDA ICE.

The end-of-life stages C2–C4 include transport of waste materials, waste processing, and final disposal. These stages were included to account for the climate impact associated with the treatment of materials at the end of the building life cycle.

3.2.3 Life Cycle Impact Assessment (LCIA)

Environmental analyses were performed in One Click LCA using the Global Warming Potential (GWP100) indicator, based on EN 15804 +A2 [40]. Greenhouse-gas emissions (CO_2e CH_4. N_2O. etc.) were characterized into CO_2-equivalents per functional unit. No weighting or normalization was applied; results are expressed directly in kg CO_2e /m^2.

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4 Results

4.1 IDA ICE- Simulation

4.1.1 Energy Performance

The electricity for heat pump operation shown in Figure 5. The BBR box shows a total usage of 77 kWh/m², while the passive box has a significantly lower value. as shown by 36 kWh/m².

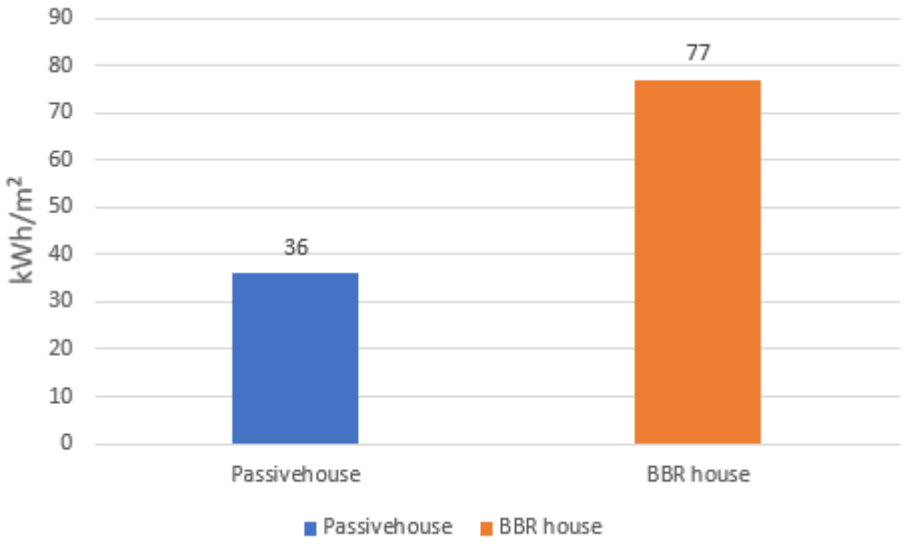


Figure 5. Electricity demand for heat pump operation in kWh/m² for both boxes.

The BBR box shows a heating demand with total usage of 231 kWh/m² as shown in Figure 6. while the passive box has a significantly lower value by 108 kWh/m².

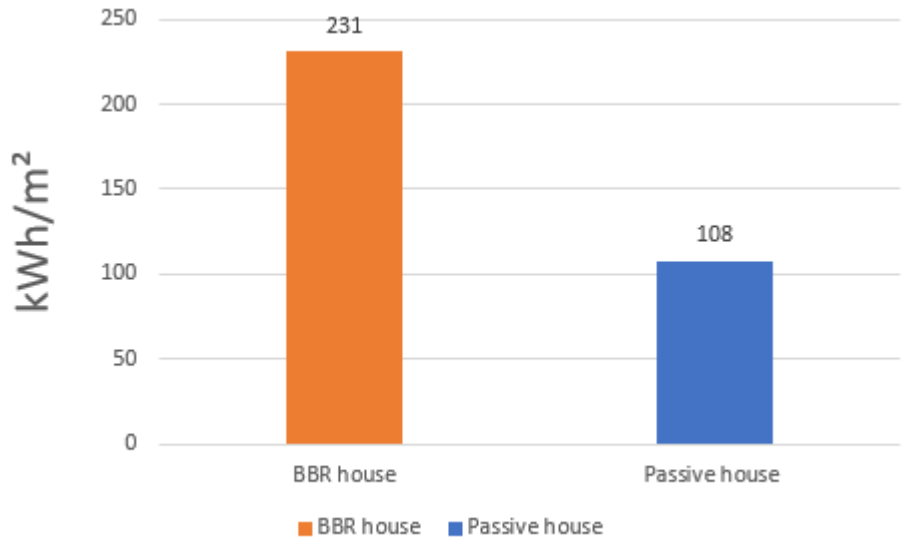


Figure 6. Heating demand in kWh/m² for both boxes.

4.1.2 Thermal Comfort

The analysis compresses data obtained during the hottest summer day, for both boxes. The mean indoor air temperature recorded in IDA ICE for Passive house box were 23.2 °C in summer (15 Jul), contrasting with 23.6 °C in summer for BBR box, see Figure 7. Those values are based on the choice of air-to-air heat pump as heating system in both boxes, with a window-to-floor area ratio of 0.6% for the passive house box and 0.8% for the BBR box and primary glazing oriented towards the north.

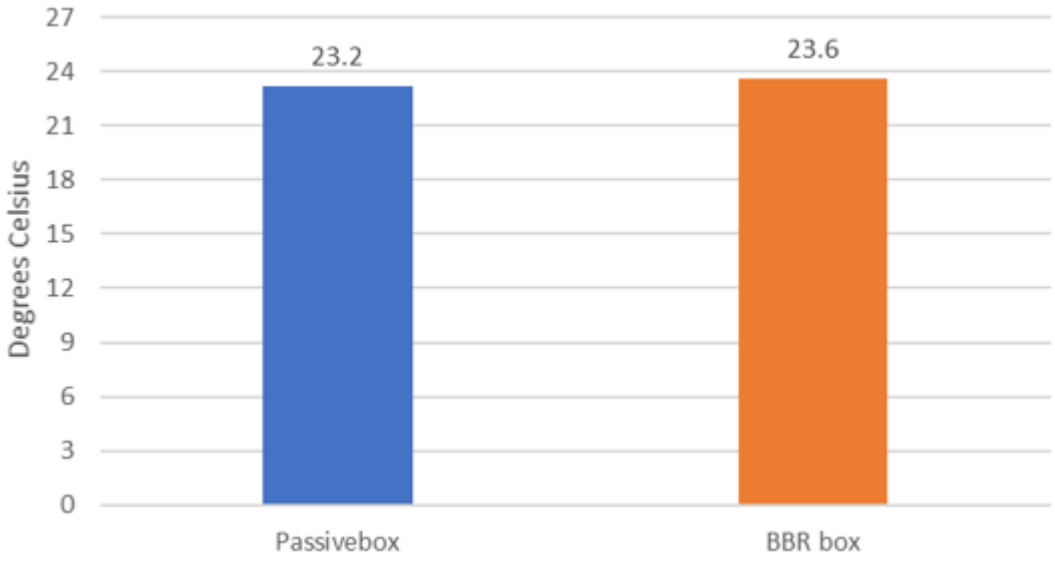


Figure 7. Maximum zone temperature under summer for passive box and BBR box.

4.2 Climate Impact/ One klick LCA

The LCA analyses from A1-A5 and the impact form insulations för both boxes are carried out in alliance with the method of the study. The results of the One Click LCA analysis show the total amount of carbon dioxide equivalents that the boxes create. Figure 8 presents information about the steps at which the carbon dioxide equivalents arise. Here it can be seen that steps A1-3 (Raw material extraction and processing phase, transport to the construction site and manufacturing) give rise to 3 360 kgCO2e for the BBR box and 2 070 kgCO_2e for the passive box. The following figure shows how much carbon dioxide equivalents come from the different materials.

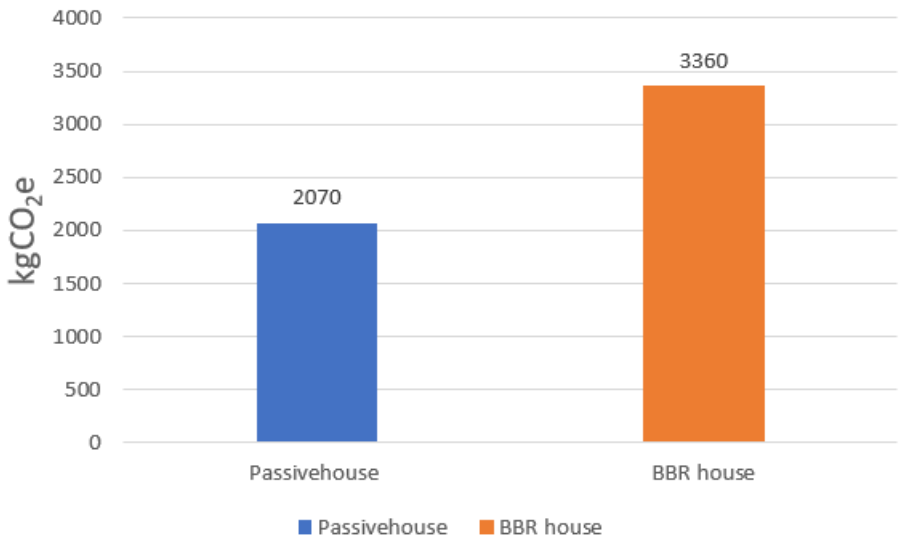


Figure 8. Carbon dioxide equivalents from respective building materials for both boxes from A1-A3.

While Figure 9 presents the impact from the transportation of building materials (A4) used in both BBR Box and passivehouse Box, the result shows that A4 gives an impact of 30 $kgCO_2e$ for the BBR box and 80 $kgCO_2e$ for the passive box. The following figure 9 shows how much carbon dioxide equivalents come from the transport.

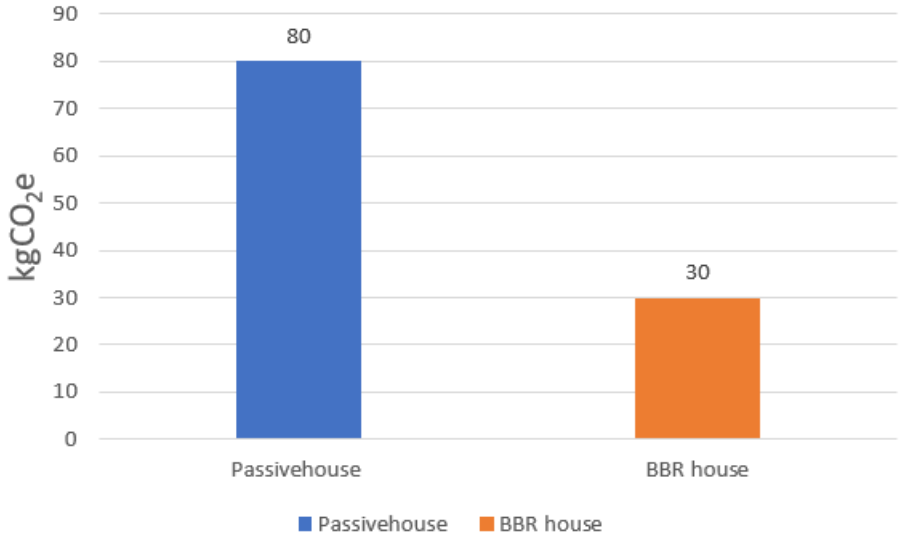


Figure 9. Carbon dioxide equivalents from the transportation (A4) of building materials for both boxes.

Figure 10 presents the impact from construction (A5). the result shows that A5 gives an impact of 450 $kgCO_2e$ for the BBR box and 370 $kgCO_2e$ for the passive box. The following figure 10 shows how much carbon dioxide equivalents come from the construction.

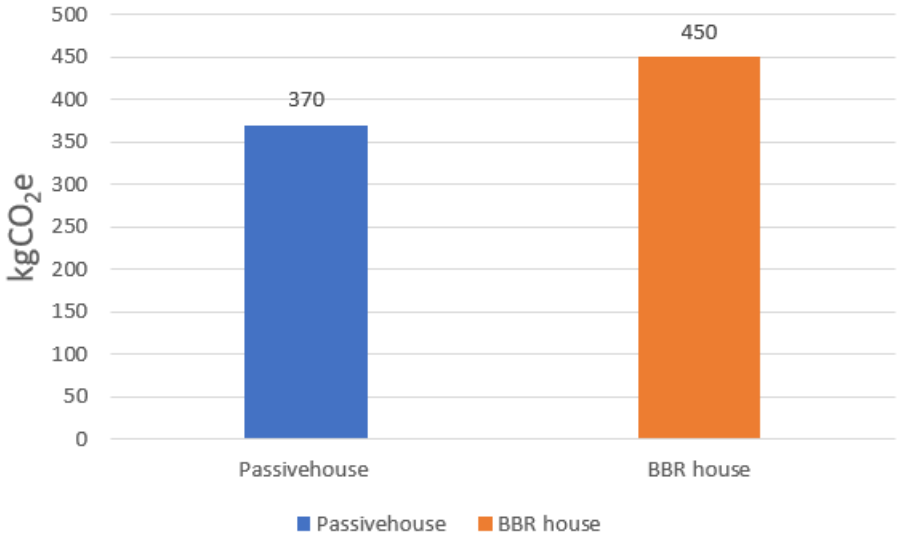


Figure 10. Carbon dioxide equivalents for the construction stage (A5) for both boxes.

Figure 11 presents the entire impact from replacement, refurbishment and operational energy use (B4-6) for both BBR box and Passive house box. The result shows that passivehouse box has a total value of 11 400 $kgCO_2e$, while BBR box has a total value of 16 790 $kgCO_2e$.

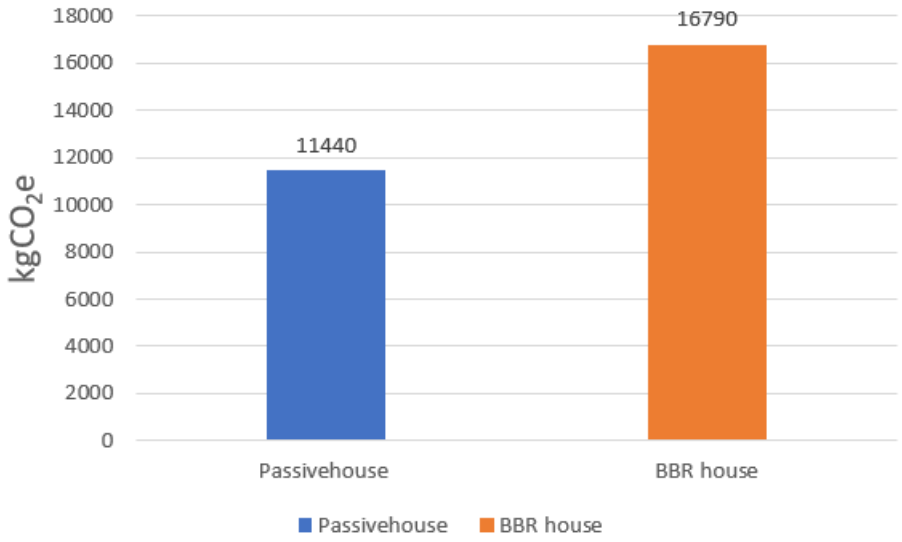


Figure 11. Carbon dioxide equivalent from replacement, refurbishment and operational energy use (B4-6).

Figure 12 presents the climate impact for the end- of- life stage (C2-4) for both BBR box and Passivehouse box. The result shows that passivehouse box has a total value of 270 $kgCO_2e$, while BBR box has a total value of 90 $kgCO_2e$.

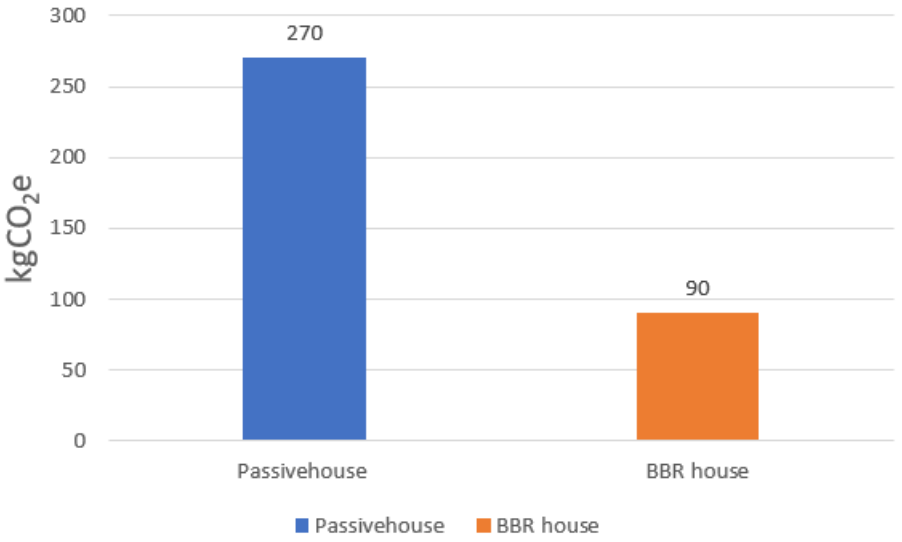


Figure 12. Carbon dioxide equivalent from life-of-end stage (C2-4).

Figure 13 presents the entire climate impact, A1-A3 (building materials stage), A4 (transportation stage), A5 (construction stage), B4-6 (replacement, refurbishment and operational energy use stage), C2-4 (life-of-end stage) for both BBR box and passivehouse box. The result shows that passivehouse box has a total value of 14 240 $kgCO_2e$, while BBR box has a total value of 20 720 $kgCO_2e$.

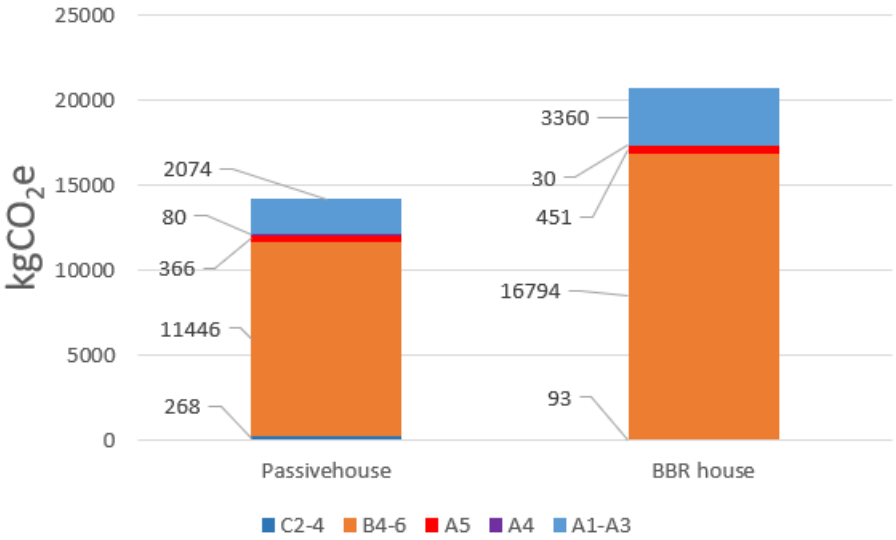


Figure 13. Total climate impact for both Passivehouse box and BBR box.

Figure 14 presents the entire climate impact from insulation for both BBR box (mineral wool) and passivehouse box (sheep wool). The result shows that passivehouse box has a total value of 496 $kgCO_2e$, while BBR box has a total value of 1 800 $kgCO_2e$.

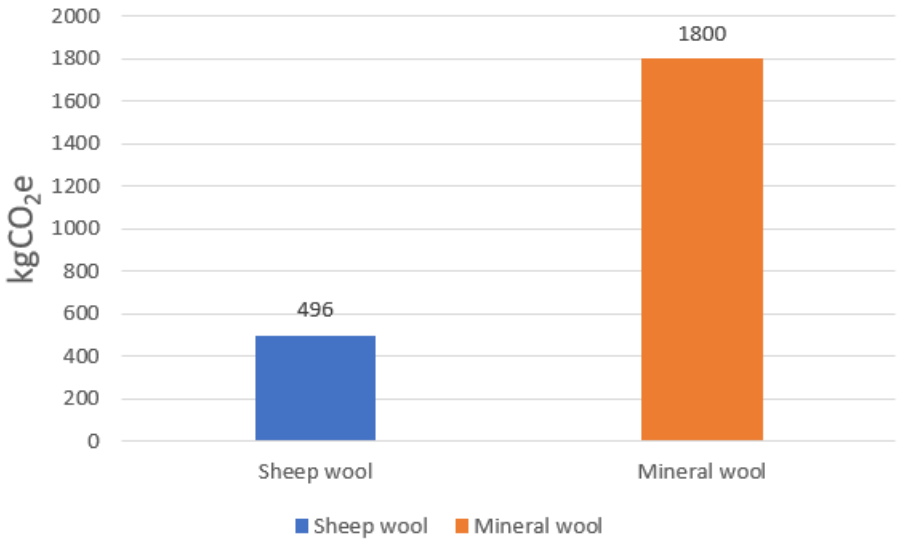


Figure 14. Carbon dioxide equivalents resulted from using sheep wool & mineral wool as insulation

5 Discussion

In this section of the report, the results and factors which could have influenced it are discussed regarding the background presented earlier. To make it structured and easier to understand, the discussion section is divided into two main parts: discussion of the results and discussion of the method.

5.1 Discussion of the results

5.1.1 IDA ICE Simulation Results

There are various factors which could have influenced the values presented in the result´s part.

First, it is important to understand that the boxes investigated in this study have different areas, which makes it unfair to reflect their energy performance, But in this case, the difference in area is not big (see table 1), which does not make a big problem when comparing the boxes, because as mentioned earlier, the study is based on the geometry of the boxes in the ICE BOX Challenge, which means that the areas used in this report are the same as in the challenge.

Secondly, the selected heat pump capacity may have influenced the energy performance outcomes for both boxes. In IDA ICE. the simulation software recommended a heat pump with a capacity in the range of 3–6 kW, based on the calculated heat losses and the prevailing climatic conditions. However, to maintain methodological consistency and ensure a fair comparison between the two cases, a considerably smaller unit with a capacity of 1.5 kW was installed in both systems. Although the boxes have limited floor areas (11.7 m^2 and 8.1 m^2), established Swedish design guidelines indicate that small buildings typically require a heating demand of approximately 40–50 W/m^2 during winter conditions [25][26]. This suggests that even compact spaces may need around 1 kW of peak heating capacity to maintain stable indoor temperatures. Additionally, it is well documented that the effective output of air-source heat pumps decreases as outdoor temperatures fall, since nominal capacities are generally measured under mild conditions (around +7 °C) and can decline significantly below (-15 °C) or lower [27][41].

In Swedish energy modelling practice, DVUT values (Design Outdoor Winter Temperature), is provided by the Swedish National Board of Housing. Building and Planning (Boverket), is used to capture the peak heating demand during the coldest periods of the year [42]. These values are essential for determining the heating capacity of the heat source required to maintain indoor comfort during winter. This consideration is particularly important for small structures such as the boxes in this study, which have a high surface-area-to-floor-area ratio. Such geometries result in higher heat losses per square meter and consequently a greater heating demand per floor are [43][44]. Taken together, these factors suggest that although the 1.5 kW heat pump was selected to enable a controlled comparison between the two box concepts, its limited capacity may have influenced the simulation results. A larger heat pump closer to the 3–6 kW range recommended by IDA ICE would likely have provided more stable indoor temperatures during colder periods. but also resulted in higher electricity use for both cases. The restricted capacity of the 1.5 kW unit is therefore a plausible explanation for why the simulated energy demand values are lower than what a fully dimensioned system would likely have produced.

Third, thermal comfort is another aspect that may have been influenced by the assumptions made in this study. In the simulations, indoor temperature was regulated solely by the heat pump, and no internal heat gains such as occupants, appliances or lighting – were included. Previous research has shown that internal loads can significantly reduce heating demand in well-insulated buildings, as they contribute a non-negligible amount of heat to the indoor environment [19]. The absence of such internal gains therefore results in colder indoor conditions and an increased heating demand to maintain thermal comfort.

On the other hand, excluding internal loads also reduces the cooling demand [17]. Who demonstrated that simulations without internal gains tend to show lower cooling energy requirements, although they may still experience overheating depending on solar exposure and envelope design. In the present study, the ideal cooling energy for both boxes was zero. The Zone Overheating result shows that the maximum/recorded zone air temperature was 23.2 °C on 15 July in the Passive House box and 23.6 °C in the BBR box. This indicates that the indoor temperature was within a generally acceptable comfort range and does not suggest a significant overheating problem at this time. According to BBR, residential buildings must be designed so that the operative temperature does not fall below 18 °C in occupied spaces [45]. Both box designs fulfil this requirement.

The Passive House box showed a slightly lower indoor air temperature, which may be explained by its better insulated envelope and more stable indoor climate. However, since the difference is small, the result does not indicate a major difference in summer thermal comfort between the two boxes. Both temperatures are within a generally acceptable indoor comfort range and do not suggest a clear overheating problem.

It is also important to note that this result is based on zone air temperature, not operative temperature. Operative temperature is an important indicator for thermal comfort because it considers both the indoor air temperature and the mean radiant temperature from surrounding surfaces. In this study, however, the operative temperature was not reported in the Zone Overheating table, where it was shown as NIL. Therefore, the summer thermal comfort discussion is based on the zone air temperature. The results showed 23.2 °C for the Passive House box and 23.6 °C for the BBR box, indicating similar indoor thermal conditions and no clear overheating problem during the analyzed summer period.

Fourth, insulation and the structure of walls have a major impact on the results of energy performance. The box built according to passive house standards has thicker layers of insulation, which means that the building envelope is highly insulated and has lower heat losses during winter. Lower heat losses mean reduced -heating demand because the energy produced is contained indoors [11]. According to the results of the study, Passive house box has energy demand of 108 kWh/m^2. while BBR box has the value 231 kWh/m^2. In contrast, the BBR house has thinner layers of insulation when compared to a passive house box, which means higher heat losses and therefore higher energy demand.

When evaluating whether the simulated energy performance aligns with established building standards, the results show a clear distinction between the two design concepts. The Passive House box achieves an energy performance of 108 kWh/m², which is significantly lower than the 231 kWh/m² (figure 6), simulated for the BBR box. The Passive House Institute defines a maximum annual heating demand of approximately 15 kWh/m² for certified Passive House buildings, together with strict requirements on airtightness, U-values, and thermal bridging [12]. Although the box in this study does not reach the 15 kWh/m² threshold, its performance remains consistent with the behavior of small Passive House-inspired constructions. Research has shown that small buildings frequently exhibit higher specific energy demand due to their large surface-area-to-volume ratio, which increases heat loss relative to treated floor area [20][21]. While carrying out the

method in Ida ICE. dimensions on the width and length where inserted, which resulted in a floor area of 11.7 m², a volume of 32.8 m³, and an envelope area of 61.8 m² for the Passive House box, and resulting in an envelope-to-floor-area ratio of approximately 5.3, Considering the box's envelope-to-floor-area ratio of 5.3, the achieved value of 36 kWh/m² remains reasonable and structurally consistent with the Passive House design principles.

In comparison, the BBR box in this study shows a simulated heating demand of 231 kWh/m². This value should not be directly compared with typical BBR regulatory energy limits, which are often expressed as primary energy numbers, EP_{pet}, in kWh/m² A_{temp} per year. For example, typical limits for electrically heated small residential buildings may fall within approximately 90–110 kWh/m² depending on climate zone [45]. However, these values refer to regulatory energy performance and not to space heating demand alone.

The high simulated heating demand can mainly be explained by the geometry of the model [20][21]. The BBR box has a floor area of approximately 8.2 m² and an envelope area of 48.3 m², giving an envelope-to-floor-area ratio of about 5.9. This means that heat losses through the envelope are large in relation to the floor area. Since the result is normalized by the small floor area, the heating demand in kWh/m² becomes high.

Furthermore, the model represents a simplified micro-scale test box rather than a full-scale dwelling. Domestic hot water, household electricity, property electricity and realistic occupancy-related internal gains are not fully represented in the same way as in a regulatory BBR assessment [18][45]. Therefore, the simulated value should be interpreted as the space heating demand of the simplified box model, while the BBR values are used only as a general reference context.

5.1.2 LCA Results

When it comes to climate impact in One click LCA, the same materials were chosen for both boxes. To make a fair comparison between both boxes, the materials were chosen with the same EPDs and from the same building materials supplier, except for the insulation. When comparing climate impact for stages A1-A3 for both boxes, the BBR box has a higher value, with a difference of 1 290 kgCO_2e. This is because of the use of materials with higher embodied emissions. such as mineral wool.
Another reason is lower effort to achieve optimized design of floor, and roof, and walls, leading to more use of material in the construction of the BBR box. When comparing climate impact for the stage A4 for both boxes, the passive house box has higher value, with a difference of 50 kgCO_2e.
A4 explains the climate impact for the transportation of building products from the manufacturer's gate to the construction site. The passive house box having higher value for A4 may be due to the transportation of sheep wool. When it comes to the construction and installation stage A5, BBR has higher value, with a difference of 80 kgCO_2e.The reason behind this difference is that BBR sets the minimum requirements for a house´ energy performance, while a passive house follows a stricter energy efficiency standard, which results in better insulation and more efficient heat recovery.

BBR has as well a higher value when evaluating the use stage B4-6, with a difference of 5 350 kgCO_2e. The difference depends on the requirements set by BBR which represent the minimum requirements for usage. When it comes to an end-of-life stage C2-4, passive house has a higher value, with a difference of 180 kgCO_2e which is not normal in most cases. This can depend on increased use of materials in a passive house in order to make it more insulated.

When choosing an EPD for sheep wool in One click LCA, there were not many Swedish building material suppliers that offer sheep wool. This means that the chosen supplier in

this case had a longer distance and therefore higher fuel consumption from trucks or trains.

The results of climate impact when comparing mineral wool with sheep wool show that mineral wool has higher climate impact with a difference of 1069.5 $kgCO_2e$.
Mineral wool has higher climate impact due to its manufacturing process, which requires more energy and high temperature. In contrast, sheep wool requires less energy because the material is natural and renewable.

Although sheep wool is a natural and renewable material and has lower climate impact, the transportation of this material results in higher climate impact. This means that it is important to consider the material and its life cycle assessment before choosing building materials.

5.2 Discussion of the method

During the literature review, no earlier studies that are based on comparing two different houses where one of them had sheep wool as insulation were found. Therefore, there was not much to start from and build on.

IDA ICE was used in this study to simulate the energy performance of the two box alternatives. A methodological limitation is that IDA ICE. like all simulation tools, is based on assumptions. input data, and simplified representations of real building operation. One of the assumptions made in this study was that both houses had no occupancy, or other internal heat gains, operating schedules, or all variations in real conditions. As a result, the simulated energy use may differ from actual building performance.

This limitation is important because the climate impact assessment in One Click LCA is partly based on the energy simulation results from IDA ICE. Therefore, uncertainties or inaccuracies in the energy model may also influence the climate impact results. The reliability of the final results thus depends partly on the accuracy of the energy simulations and the assumptions used in the IDA ICE model.

The investigated boxes were not intended for residential use. If the boxes had been used as dwellings, internal heat gains from occupants would have affected both indoor temperature and energy demand. During the heating season, such heat gains could have reduced the need for space heating. However, during warmer periods, occupant-related heat gains could also have increased indoor temperatures and worsened thermal comfort or increased cooling demand. This means that occupancy would have influenced the results differently depending on the season.

Another methodological limitation concerns the choice of EPDs in One Click LCA. In this study, EPDs were mainly selected based on Swedish production and low reported climate impact values. However, no systematic review was made of the verification dates of the EPDs or whether newer versions were available. This means that some of the EPDs used may not have been the most up-to-date versions, which could have affected the climate impact results. In addition, the study has not conducted any background on the chosen manufacturers and their production processes and their expertise in this field.

During the phase of background research, there were a few earlier studies which investigated the impact of LCC while comparing houses build with different standards. In the report *Energi- och kostnadsjämförelse hos BBR och passiva flerbostadshus* [8] investigations on LCC for an existing apartment building in Stockholm were made, and how it would have differed if it was constructed in accordance with passive house standards. The report shows that LCC is an important factor when evaluating a building performance over time. The results of LCC in the report shows that higher initial costs are balanced out by lower operational costs.

If LCC was included in this study, the results may have been affected by providing a more detailed comparison on costs savings between both boxes. Sustainability takes into consideration the economic viability. Including LCC in the study would have allowed an evaluation of long- term costs such as investment, operation, and maintenance for both boxes, which are important to determine if a solution, such as building according to the passive house standard, is sustainable in practice. This evaluation would have made an important basis for decision making and implementation of solutions that can be environmentally beneficial but economically unbeneficial. As mentioned earlier in the report, designing according to passive house standard means high initial costs, but is more efficient than BBR houses in long terms. Therefore. it is hard to fully evaluate the overall sustainability of both boxes since the economic aspects are also an important factor.

6 Conclusions

The purpose of the thesis is to study how the design of the two buildings used in the Ice Box Challenge in Borlänge, one following BBR standards and the other passive house standards, affect energy demand, thermal comfort and the climate. The aim is to investigate both buildings' impact on the climate according to the required energy for them. The conclusions of this thesis answer.

How heating demand differs in both studied boxes?
The results for that question show that heating demand for the box following BBR standards had higher value in comparison with the box following passive house standard. The heating for BBR box was 231 kWh/m², and for the passive house was 108 kWh/m².

How does the climate impact differ in comparison between BBR box and passive box in the context of the Ice Box Challenge?
The results for those questions show that the climate impact simulations run in One click LCA differ for the studied boxes:

A1-A3, The BBR box has a climate impact of around 3 360 $kgCO_2e$. (16%) while the Passive house box has around 2 070 $kgCO_2e$ (15%), see figure 8. This means that the BBR box has a higher climate impact with a difference of around 1 290 $kgCO_2e$.

A4, BBR box has a value of around 30 $kgCO_2e$ (0.1%) while Passive house box has a value of around 80 $kgCO_2e$ (0.5%), see figure 9. The passive house box has a higher value than the BBR box. with a difference of around 50 $kgCO_2e$.

A5, BBR box has a value of around 450 $kgCO_2e$ (2%) while passive house box 370 $kgCO_2e$ (3%) BBR box has a higher climate impact. with a difference of 90 $kgCO_2e$, see figure 10.

Insulation, mineral wool has a climate impact of around 1800 $kgCO_2e$ (8.7%), while sheep wool has around 496 $kgCO_2e$ (3.5%). This means mineral wool has a higher climate impact, with a difference of around 1069.0 $kgCO_2e/ m^3$. see figure 14.
B4-6, BBR box has a value of around 16 790 $kgCO_2e$ (81%), while passive house box 11 440 $kgCO_2e$ (80%) BBR box has a higher climate impact, with a difference of 5 350 $kgCO_2e$, see figure 11.

C2-4, passive house box has a value of around 270 $kgCO_2e$ (1.3%), while BBR box 90 $kgCO_2e$ (0.4%). Passive house box has a higher climate impact, with a difference of 180 $kgCO_2e$, see figure 12.

The total climate impact for all stages for the BBR box is around 20 720 $kgCO_2e$, and for the passive house box is around 14 240 $kgCO_2e$, see figure 13. The results show that BBR has higher climate impact compared to the passive house box, with a total difference of around 6 480 $kgCO_2e$ for A1-A5. B4-6 and C2-4. However, there is a possibility that the comparison is affected by the different in the quantities of the materials such as insulation and the total area. The climate impact may have differed if both boxes were modeled with identical material specifications.

6.1Future Work

The models used represent a limited area and are not based on actual buildings, which means that the results are more theoretical than practical. In this project, there are also no people living in the boxes, which means that the models need to include internal loads from people, lighting, and household appliances to reflect real-world operation more accurately.

In addition. a cooling source should be included. as the results from IDA ICE showed that no cooling source was needed, but this may be due to the lack of internal loads, and if all systems are included. the results may change.

During the literature review, studies on the use of sheep wool insulation for buildings were not found, which means less information on its advantages/disadvantages for energy demand and climate impact. There is a gap when it comes to sheep wool, and that needs to be investigated. An example of future work is to investigate the effects of sheep wool in a real case house to connect it to reality and understand if it really has benefits and should be used more. Another aspect that can be studied in the future is investigating LCC for buildings that are insulated with sheep wool to understand if the material contributes to circular economy.

IDA ICE and One Click LCA are used in research and the construction industry. If the correct input data is given, the results are reliable; however, in this study, the model has been simplified by the absence of individuals, cooling needs or different systems. Future results can be compared with other programs or measurements from real buildings to increase reliability.

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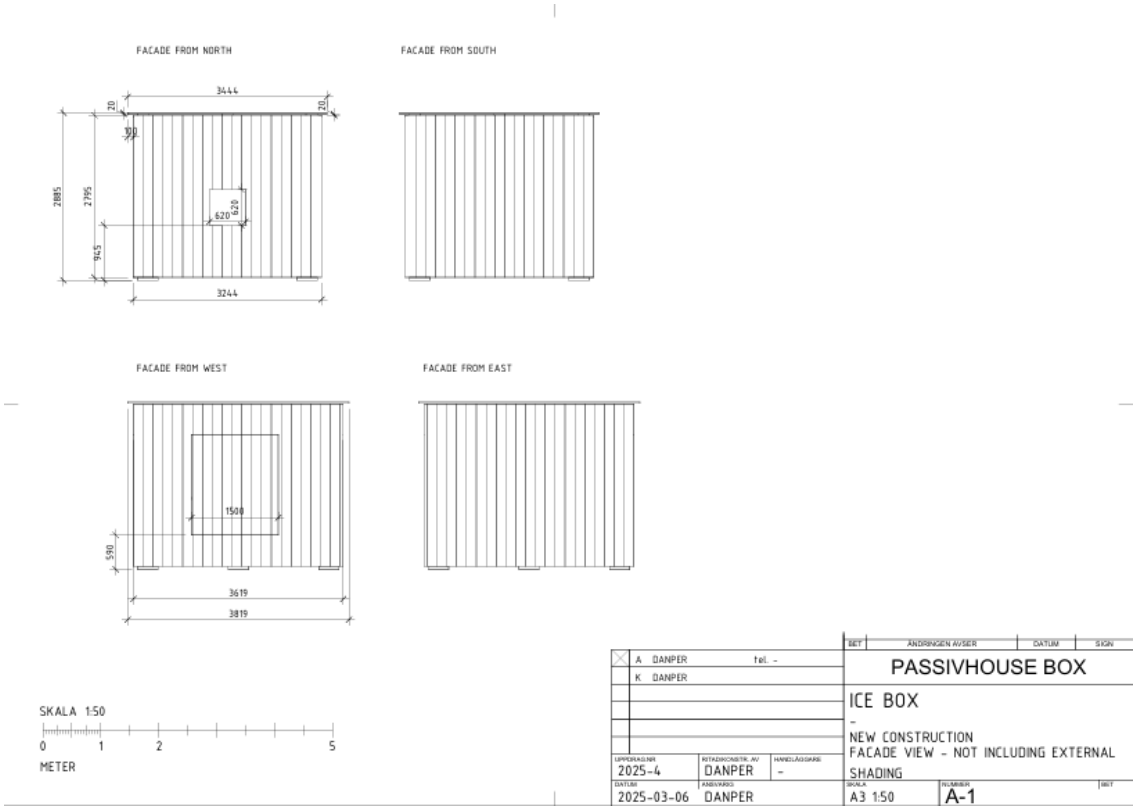
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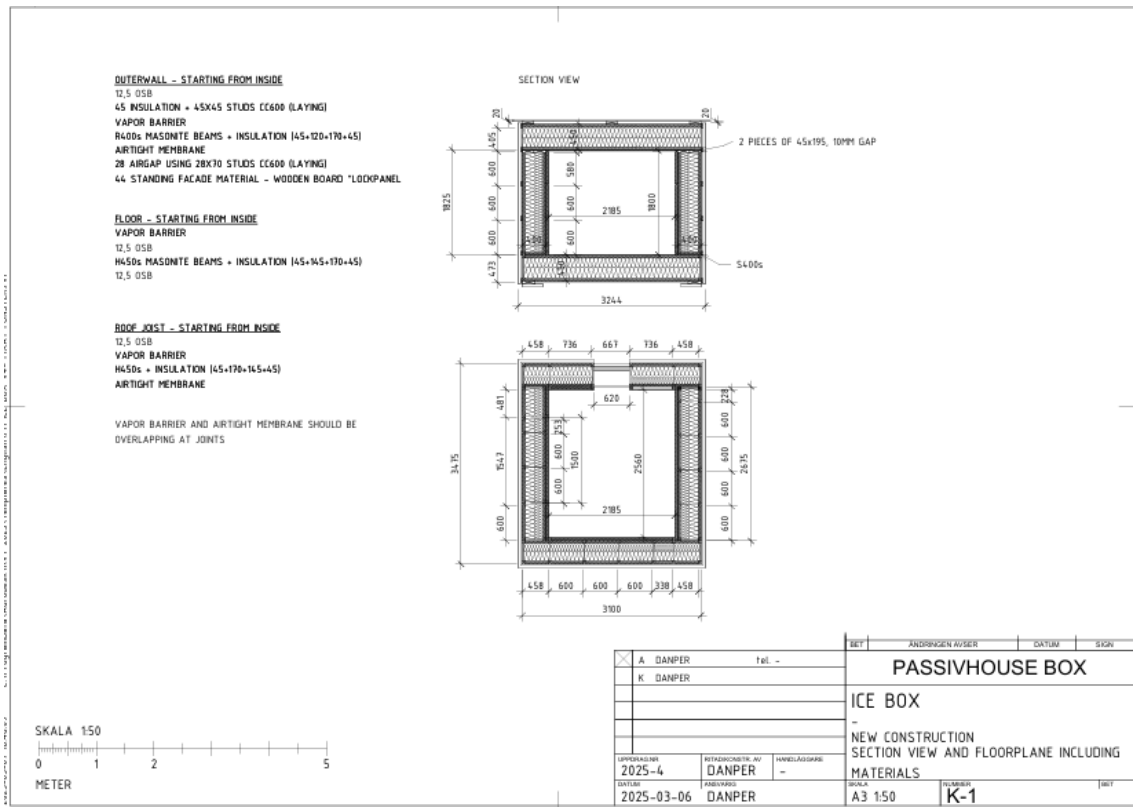
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Appendices

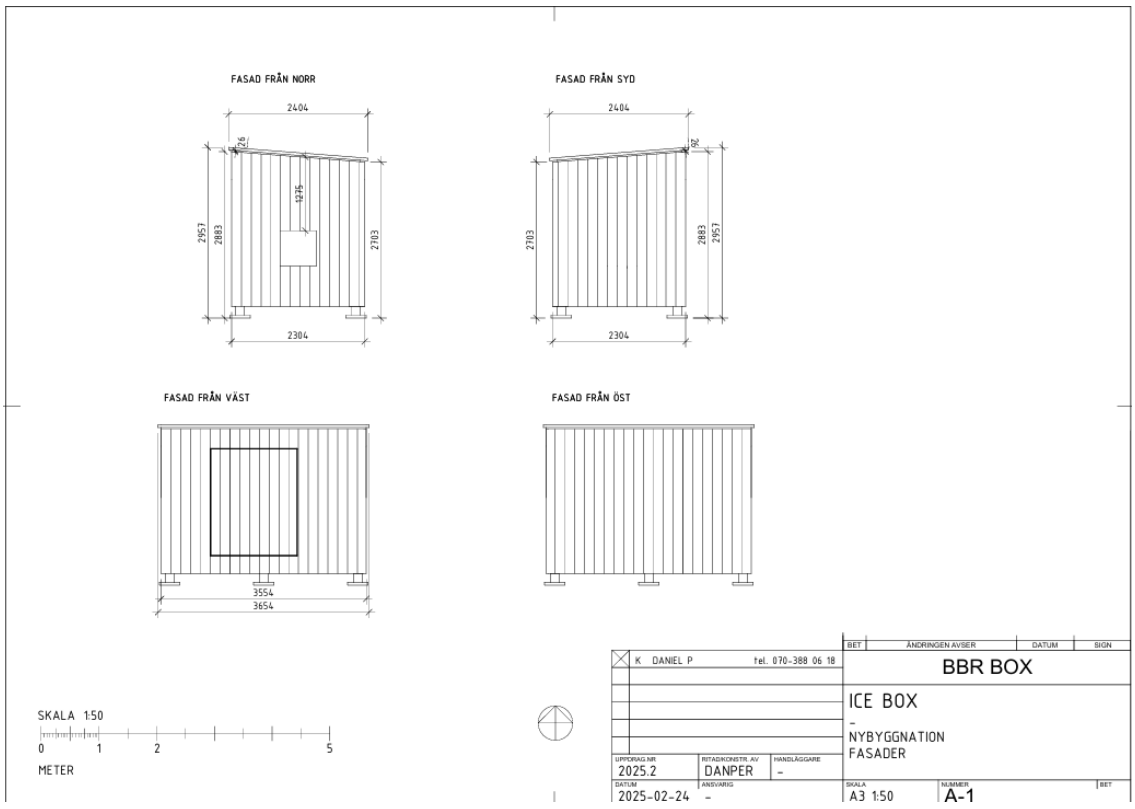
1. Drawing

1.1 Passive Box Drawing

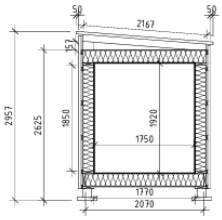




1.2 BBR Box Drawing



SEKTIONSVY FRÅN NORR

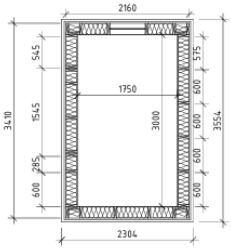


YV - INSIDA UT
10 OSB
ÅNGSPÄRR
45X195 C24 REGEL
VINDSKYDDSDUK
28X70 LUFTSPALT
44 LOCKPANEL

TAK - INSIDA UT
45X200 C24 REGEL + ISOLERING
ÅNGSPÄRR
10 OSB

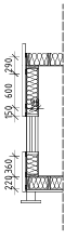
GOLV - INSDIA UT
TÄTSKIKT
10 OSB
45X220 C24 REGEL + ISOLERING
10 OSB

FUNDAMENT - FRÅN MARK
MARKSTEN
150X150 BETONGKUB
SYLLISOLERING



PLANRITNING

ÅNGSPÄRR OCH VINDSKYDDSDUK VIKS IN VID ÖPPNINGAR



FÖNSTER DETALJ

BET	ÄNDRINGEN AVSER	DATUM	SIGN

K DANIEL P tel. 070-388 06 18

BBR BOX

ICE BOX
-
NYBYGGNATION
MATERIAL

UPPDRAG NR	RITAD/KONSTR. AV	HANDLÄGGARE
2025.2	DANPER	-
DATUM	ANSVARIG	

SKALA	NUMMER	BET
A3 1:50	K-02	